Reading the Prompt, Not the Economics

Structural Estimation of an AI Model's Risk Preferences

A progress report on Phases 1 and 2, with the Phase 3 plan

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Across two phases, Claude's risky choices track the surface features of a prompt, the probability figure and the word "loss," more than the underlying economics.

This project treats a large language model as an economic agent and recovers its preference primitives from its choices. Two phases are complete. Each elicits roughly a thousand to two-and-a-half thousand controlled binary lottery choices from a pinned model snapshot, then fits a structural discrete-choice model to the result. Phase 1 asks how the model weighs risk. Phase 2 asks whether the wording of a choice, holding the money fixed, moves the answer. Both return sharp, quantified, human-comparable results. Phase 3, planned below, asks where the biases come from.

Phase 1: How Claude Weighs Risk

`claude-haiku-4-5-20251001` · 1,200 controlled choices · forced binary choice between a sure amount and a two-outcome gamble

2.69	0.59	22.8 vs. 5.0
probability-weighting curve; humans about 0.7 and bent the opposite way	risk-aversion once weighting is controlled; squarely human-plausible	how strongly the win-probability drives choice versus the payoff size

A first, simple risk model fit badly. A textbook constant-relative-risk-aversion (CRRA) expected utility model implied a risk-aversion of 0.19 and could not reproduce the choices. The diagnostic showed why. As the win probability rose from low to high, gamble-taking climbed from 8% to 86%, while the gambles' actual expected value stayed roughly flat. Claude was pricing the probability and nearly ignoring the dollar amounts.

An extended model, adding a probability-weighting curve and a position term, fit far better: a likelihood-ratio chi-square of 617 on 2 degrees of freedom. Its probability-weighting curve is an extreme S-shape at 2.69, the opposite of the human inverse-S. Claude underweights long shots instead of overweighting them, so it will refuse a long shot at almost any prize. Once that weighting is controlled, its risk-aversion of 0.59 is ordinary and human-plausible; the earlier 0.19 was an artifact of the wrong model. Choice noise roughly halved, from 0.26 to 0.10, once the model was right.

Key finding

Claude prices the probability of winning and nearly ignores the size of the prize. Its probability-weighting curve is an extreme S-shape pointed opposite to the human one: it underweights long shots rather than overweighting them, and will pass on a high-value long shot at almost any prize.

M2 parameter estimates. CRRA value, Prelec probability weighting, and a position term. $n = 1{,}200$.

Parameter	Estimate (se)	Interpretation
r (curvature)	0.592 (0.028)	Risk aversion, human-plausible range
γ (Prelec)	2.693 (0.166)	Extreme S-shape; long shots underweighted
μ (Fechner noise)	0.104 (0.008)	Choice noise, less than half of the baseline
δ (position)	1.257 (0.160)	About 3.5× odds pull toward the option listed second

Phase 2: Framing and the Fear of Losses

`claude-haiku-4-5-20251001` · 2,480 controlled choices · one terminal-wealth lottery, shown in three logically identical frames

40% → 10%
gamble-taking on the same lottery when the downside is worded as a loss instead of getting nothing

3.23
estimated fear-of-losses, above the classic human figure of about 2.25

100%
a built-in control, where one option was plainly larger, answered correctly every time

The same terminal-wealth lottery was shown three ways that a rational agent must treat identically: as pure gains, in neutral wording, and as a mix where the downside is called a loss. Gamble-taking was 40% in the gain wording, 30% neutral, and 10% when the downside was a loss, a gap of about 0.30 with a p-value on the order of 10^{-47}. Holding each individual lottery fixed, 88% of them show the effect, so it is not composition. This violates frame invariance: identical money, different words, different choice.

The structural estimate puts the fear-of-losses at 3.23, above the canonical human figure of about 2.25. The dominant-choice control, a certain amount versus a strictly larger certain amount, was answered correctly 100% of the time. Claude reads dollar magnitudes perfectly well when no probability is involved; the trouble is specific to choices under risk, and here, specific to the word "loss."

Key finding

Two lotteries with byte-for-byte identical terminal wealth draw 40% versus 10% gamble-taking based only on whether the downside is called a loss. The estimated loss-aversion coefficient, 3.23, exceeds the canonical human 2.25. Wording moves the choice; the money does not change.

An honest identification limit

The data cannot fully separate "the reference point moved" from "an extreme fear of losses." The mixed-frame suppression admits two nearly observationally equivalent readings, so the clean headline is the assumption-free frame gap, and the fear-of-losses figure of 3.23 is reported under the standard

Structural estimates under the stated-reference assumption ($\phi = 1$, $n = 2{,}480$).

Parameter	Estimate (se)	Interpretation
λ (loss aversion)	3.234 (0.171)	Losses loom about 3.2× larger than equal gains
α (curvature)	0.503 (0.024)	Diminishing sensitivity, human-plausible
γ (Prelec)	1.516 (0.059)	S-shape; long shots underweighted
μ (Fechner noise)	0.131 (0.009)	Choice noise
δ (position)	−0.363 (0.104)	Pull away from the gamble when listed second

stated-reference assumption. One further caution: the position term flipped sign between phases, from +1.26 in Phase 1 to −0.36 here, so position bias is a design artifact rather than a stable trait of the model.

Synthesis: Two Phases, One Thread

The emerging portrait

Claude's choices under risk are governed by salient surface features of the prompt, the probability number in Phase 1 and the loss wording in Phase 2, more than by an integrated computation over final wealth. It reads magnitudes fine when nothing distracts it, as the 100% control shows. So this is selective attention, not an inability to do arithmetic.

Phase 3: Where Do the Biases Come From?

This is the payoff the whole project was built toward. Phases 1 and 2 measured the biases. Phase 3 asks the attribution question: are these distortions written into the model during pretraining, or installed by the final alignment step?

Method. Run the identical Phase 1 and Phase 2 instruments on matched pairs of open models, one *before* its final training step and one *after*, that is, base versus instruct. Read each choice probability directly from the model's own token probabilities rather than from sampled text. If a bias appears only in the after-training model, the training instilled it. A format-control arm, running the after-training model in both prompt formats, rules out a wording confound.

Models. Qwen2.5-7B goes first, then Llama-3.1-8B, with OLMo-2 as a stretch. OLMo-2 publishes its staged checkpoints, base, then supervised, then preference-tuned, which would let the bias be pinned to a *specific* training stage.

The question Phase 3 answers

Is a model's economic character learned in pretraining, or written in by the alignment step at the end?

Status and practicalities. The pipeline is built and already validated on simulated data, where it recovers known values near-perfectly. It runs on a free Colab GPU, because no hosted service serves the needed before-training models with token probabilities.

Estimates are properties of the pinned snapshots under this unincentivized forced-choice protocol; they are not claimed as universal traits of language models.